

Opportunities For 3D PRINTING BUSINESS In India



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3D printers are emerging as a key growing printing technology in the global landscape. In India, the market for 3D printers is at its nascent stage. However, it offers huge growth opportunities in the coming years

3D printing helps the industry think faster, innovate better and broaden up the horizons of manufacturing. It is an example of technology and manufacturing working together to become more flexible and scalable, ensuring enhanced efficiency and quality.

There has been a gradual shift from conventional manufacturing techniques to new and emerging technologies that provide higher precision and resolution, thus speeding up the manufacturing process.

3D printer software enable users to create digital models from which physical objects are made. Primary benefit of this technology is its ability to manufacture customised parts as needed without costly machinery or dies for casting.

3D printing is a rapidly growing industry. It has enhanced its value in diverse sectors such as automotive, healthcare, aerospace and defence (A&D), consumer products, industrial products, education and research.

Advantages of 3D printing

3D printing technology helps industries streamline design processes and lower production costs. Today, manufacturers across a broad spectrum of industries, including automotive, aerospace, dental, discrete, high-tech and medical products are all actively piloting and using 3D printing technologies. Prototyping continues to be the dominant reason why

3D printing: At a glance

3D printing is a process through which 3D objects are built by stacking successive layers of raw materials as per desired form. Deposition of layers is controlled by computer-aided design (CAD) software as per the following parameters:

1. Virtual 3D blueprint
2. Technical specifications needed for fabrication (measurements such as length, width and height) of each layer

As per specifications, the software slices the virtual model into several layers. Then, a string of codes is sent to the 3D printer defining how to stack the layers of raw material. Thickness of a layer can be as low as one-thousandth of a metre. Thus, 3D printing is recognised as an extremely high-precision manufacturing procedure. It is predominantly used for producing functional parts and models/prototypes for various applications in sectors including automotive, A&D, healthcare and consumer durable end markets.

enterprises adopt 3D printing, providing the opportunity to speed up new product development and time-to-market.

3D printers facilitate manufacturing by:

1. Helping re-engineer lighter and better designed versions of existing parts
2. Being cost-effective for design changes
3. Allowing low-risk manufacturing for market testing
4. Promoting customisation and personalised products
5. Improving accuracy
6. Decreasing expensive mistakes
7. Promoting collaborative design between departments
8. Highlighting costly design flaws
9. Improving customer and designer involvement and, hence, overall satisfaction
10. Speeding up decision-making

Moreover, 3D printing methods also ensure substantial savings on tooling in comparison with traditional manufacturing technologies, such as injection moulding, computer numerical control (CNC) method and vacuum casting.

Market opportunities

3D printing is emerging as the fastest growing technology worldwide. According to

3D printer in action

